

### 1. getElementById

Purpose: Targets a specific, unique element using its id attribute.

Return Type: It returns a single DOM element. If no match is found, it returns null.

Key Detail: Since IDs must be unique on a page, this is the most efficient and fastest way to select a single element. It can only be called on the document object.

### 2. getElementsByClassName

Purpose: Selects all elements that have a specific class name.

Return Type: It returns a live HTMLCollection.

Key Detail: "Live" means if you add or remove elements with that class later in your code, the collection updates automatically. To manipulate these elements, you must loop through the collection or access them via index (e.g., elements[0]).

### 3. getElementsByTagName

Purpose: Selects all elements of a specific HTML tag type (like p, div, or button).

Return Type: Like the class name method, it returns a live HTMLCollection.

Key Detail: This is useful for global changes, such as applying a style to every paragraph on a page. You can call this on a specific parent element to find only its descendants (e.g., myDiv.getElementsByTagName('span')).

### Method Selection Criteria Return Value Scope

getElementById Unique id Single Element (or null) document only

getElementsByClassName One or more class Live HTMLCollection document or specific element

getElementsByTagName HTML tag (e.g., div) Live HTMLCollection document or specific element